REVISION FOR FINAL EXAM

SEM 1 20232024

ARTIFICIAL INTELLIGENCE

SECJ3553

PART A – OBJECTIVE QUESTIONS

[20 marks]

Part A consists of 20 objective questions. Each question carries 1 mark. Choose the <u>correct answer</u> and write your answer in the test booklet.

1.What is the evaluation function in A* approach? $f(n) = g(n) + h(n)$

a) Heuristic function

b) Path cost from start node to current node

c) Path cost from start node to current node + Heuristic cost

d) Average of Path cost from start node to current node and Heuristic cost

2. Heuristic function h(n) is ________

a) Lowest path cost

b) Cheapest path from root to goal node

c) Estimated cost of cheapest path from root to goal node

d) Average path cost

3. A* algorithm is based on ___________

a) Breadth-First-Search

b) Depth-First –Search

c) Best-First-Search

d) Hill climbing

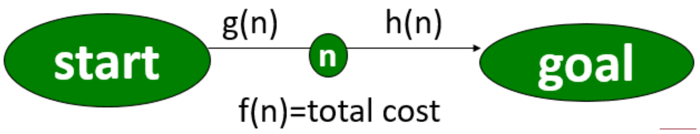
CH 4B Search Algorithms -HEURISTIC EVALUATION

1. Which evaluation criterias can be applied on a heuristic function?

   I. Informedness
   II. Admissibility
   III. Monotonocity
   IV. Consistency

   A. I and IV
   B. I, II, and III
   C. I, II, III and IV
   D. I and III

2. Which statements are TRUE about the Figure 1 below



   I. $f(n) = g(n) + h(n)$ estimates the total cost of path from start state through n to the goal state
   II. g(n) is the actual distance from n to start state
   III. For 8 puzzle problem, h(n) could be no of tiles in wrong position
   IV. g(n) is the actual distance from n to goal state

   A. I and IV
   B. I, II, and III
   C. I, II, III and IV
   D. I and III

3.. Which statements are TRUE about A* algorithm

   I. A* algorithm is optimal if its h(n) is admissible
   II. All A* algorithms are admissible
   III. Monotone or consistent heuristic is an admissible A* algorithm
   IV. Not all admissible A* algorithms are monotone or consistent

   A. I and IV
   B. I, II, and III
   C. I, II, III and IV
   D. I and III

CH 4C Problem Solving with search

1. Alpha Beta pruning algorithm follows _______search and allows pruning of branches in __________fashion.

A. Breadth-first, backtracking

B. Depth-first, backtracking

C. Best-first, forward chaining

D. Best-first, backtracking

2. Which of the following statement is TRUE about Alpha Beta procedure?

A. Alpha Beta pruning algorithm follows best-first search strategy.

B. *n-ply-look-ahead* can be applied in a complicated games where it is impossible to expand the state graph out to leaf nodes.

C. In alpha beta pruning, Max will update β value whilst Min will update α value.

D. In alpha beta procedure, two values are created, i.e α associates with MAX-never decrease and β associates with MIN-never increase

3. Which of the following statement is FALSE about Alpha Beta rules?

A. While backtracking, the node values are passed to the parent

B. While backtracking, the α and β values are passed to the parent

C. MAX only update α, MIN only update β

D. Condition to fulfill when pruning is : α>= β

4. Which of the following statement is FALSE about Alpha Beta procedure?

A. It provides faster searching because it can prune a leaf or entire branch at any depth of tree.

B. It improves search efficiency in two-person games compared to minimax that always pursues all branches in state space

C. Alpha (α) is the highest value found along the path of MAX and start at ∞

D. Alpha (α) is the highest value found along the path of MAX and start at -∞

MINIMAX

1. According to the rule of MINIMAX, if the parent state is at minimum, the __________ value from its children will be assigned to the parent.

A. minimum
B. maximum
C. 0
D. 1

CH5 Search Planning and Control

1. The components in production systems includes the following

I. Production Rules

II. Working memory

III. Recognize-act control cycle

IV. Conflict Routine

A. I, II, III

B. I, II, IV

C. II, III, IV

D. I, III, IV

3. ________ is an inference technique that uses IF-THEN rules to repetitively breaks a goal into smaller sub-goals that are easier to prove.

A. Forward chaining

B. Backward chaining

C. Heuristic search

D. Exhaustive search

CH 6 Advanced Artificial Intelligence

1. AI Agent use ____________ and ____________ to perceive and act based on the environment.

A. Sensors, Actuators
B. Sensors, Performance
C. Sensors, Action
D. Sensors, Goal

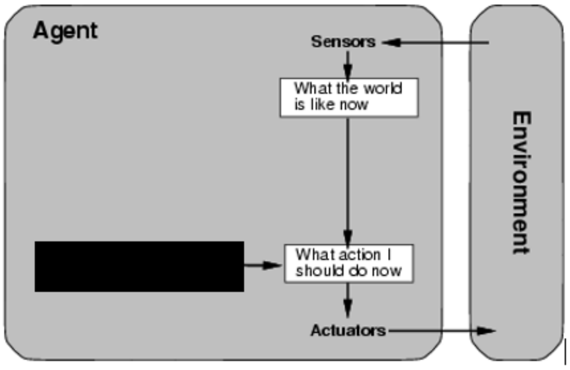


Figure 1: AI Agent

2. What is the suitable rule to be implemented by an AI agent in Figure 1?

A. Simple action rule
B. Condition action rule
C. Performance action rule
D. Goal action rule

3. What is the type of intelligent agent which relies on current conditions without referring to the historical data?

A. Utility-based
B. Goal-based
C. Simple reflex
D. Learning

4. Which is best to describe the purpose of an intelligent agent?

A. To respond based on the surrounding environments.
B. To facilitate human-to-computer interaction.

C. To determine whether an answer is right or wrong.
D. To acquire vast amounts of searchable data.

5. An agent must perceive data using ______ in order to operate in the environment.
A. Actuators
B. Sensors
C. Factors
D. Refactors

CH7 Machine Learning

1. Which one of the following is **TRUE** about clustering?
I. It follows trial and error fashion
II. It uses similarity measures to group the data.
III. It is a reward-based technique
IV. It learns from unlabelled data
A. I and IV
B. I and II
C. II and IV
D. I and III

2. Which learning technique should be used to train an automotive robot painter?
A. Classification
B. Regression
C. Clustering
D. Reinforcement learning

	Predicted: 0 N	Predicted: 1 P	
Actual: 0 F	8 TN	2	FP
Actual: 1 T	0 FN	1	TP

Recall = TP/(TP + FN)
= 1/(1 + 0)
= 1

3.What is the correct interpretation of the above confusion matrix?

Precision = TP/(TP+ FP)
= 1/(1+2)
= 1/3

A. The value of type I error is 0. 2 FP

B. The value of type II error is 1. 0 FN

C. The recall value is 1. ✓

D. The precision value is 1.

PART B – STRUCTURED

CH 4B Search Algorithms

1. An A* search is used to solve the 8-puzzle problem game with heuristic h(n) as the number of tiles out of place. [3m]

   a) The heuristic h(n) is admissible when h(n) <= h*(n), explain what does this means?[1m]

   b) Explain how is h(n) an informed heuristic in 8 puzzle game [2m]

QUESTION 2- CH5 Search Planning and Control

1. Let,

A = the ignition key is on

B = the engine will not start

C = headlights work

D = the voltage test on the ignition switch shows 1 to 6 volts

(A, B, C and D are the FACTS in the Initial Database)

E = the starting system (including battery) is faulty

G = starter is faulty

I = battery is dead

F = wiring between the ignition and the solenoid is good

H = replace the ignition switch

(E, F, G, H, I are the FACTS that can inferred from the FACTS in the Initial Database)

Rules are given below and they are in the order of R1, R2, R3, R4 and R5:

R1: $A \wedge B \rightarrow E$ ✓

R2: $E \wedge C \rightarrow G$ ✓

R3: $E \wedge \sim C \rightarrow I$

R4: $D \rightarrow F$ ✓

R5: $F \rightarrow H$ ✓

Consider conflict set as a queue and rule must be fired from front of the queue

(a) Use the forward chaining to infer all the possible FACTS from Initial Database (10M)

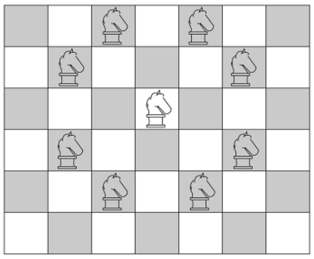
Iteration #	Working Memory	Conflict Set	Rule Fired
1	A, B, C, D	R1, R4	R1
2	A, B, C, D, E	R1, R4, R2	R4
3	A, B, C, D, E, F	R1, R4, R2, R5	R2
4	A, B, C, D, E, F, G	R1, R4, R2, R5	R5
5	A, B, C, D, E, F, G, H	R1, R4, R2, R5	Halt

(b) Use backward chaining to infer (H). (Hint: Stop the iteration when sub-goal is able to reach FACT in the Initial Database (either A, B, C or D) (5M)

Iteration #	Working Memory	Conflict Set	Rule Fired
1	H	R5	R5
2	H, F	R5, R4	R4
3	H, F, D	R5, R4	Halt

(a) **Given a problem of knight's tour in Figure 1 and production rules for the 3 X 3 knight problem in Table 1, please state a production system that shows a knight make a move from 1 to 4. Show your answer as shown in Table 4. (5M)**

1.



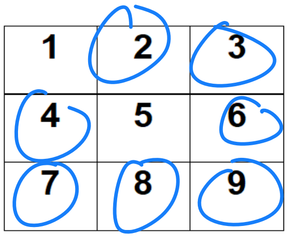


Table 1

RULE #	CONDITION		ACTION
1	knight on square 1	→	Move knight to square 8
2	knight on square 1	→	Move knight to square 6
3	knight on square 2	→	Move knight to square 9
4	knight on square 2	→	Move knight to square 7
5	knight on square 3	→	Move knight to square 4
6	knight on square 3	→	Move knight to square 8
7	knight on square 4	→	Move knight to square 9
8	knight on square 4	→	Move knight to square 3
9	knight on square 6	→	Move knight to square 1
10	knight on square 6	→	Move knight to square 7
11	knight on square 7	→	Move knight to square 2
12	knight on square 7	→	Move knight to square 6
13	knight on square 8	→	Move knight to square 3
14	knight on square 8	→	Move knight to square 1
15	knight on square 9	→	Move knight to square 2
16	knight on square 9	→	Move knight to square 4

Answer:

Iteration	Working Memory		Conflict Set (Rule No)	Fire Rule
	Current	Goal		
0	1	4	R1, R2	R1
1	1,8	4	R1, R2, R13	R13
2	1,8,3	4	R1, R2, R13, R5	R5
3	1,8,3,4	4	R1, R2, R13, R5	Halt

QUESTION 3 - CH 6 Advanced Artificial Intelligence

CHAPTER 6: ADVANCED ARTIFICIAL INTELLIGENCE:

A self-driving car, also known as an autonomous vehicle (AV), driverless car, or robotic car (robo-car), a car incorporating vehicular automation. That is, a ground vehicle which capable of sensing its environment and moving safely with less or without human input.

Table 1: Autonomous Vehicles Agents PEAS

EXAMPLE	PERFORMANCE / GOAL	ENVIRONMENT	SENSOR / PERCEPT	ACTUATOR / ACTION
Autonomous vehicle	A	B	C	D

1. Referring to Table 1, what is the performance (A) I for the case study? (2 marks)
   a. Able to stop when detect obstacles
   b. Accelerate on the normal driving
   c. Decide the shortest path

2. Referring to Table 1, what is the performances (A) II for the case study? (2 marks)
   a. Monitor driving environments
   b. Able to understand the signage
   c. Use the traffic information and update current traffic

3. Referring to Table 2, what is the Environment (B) for the case study? (2 marks)

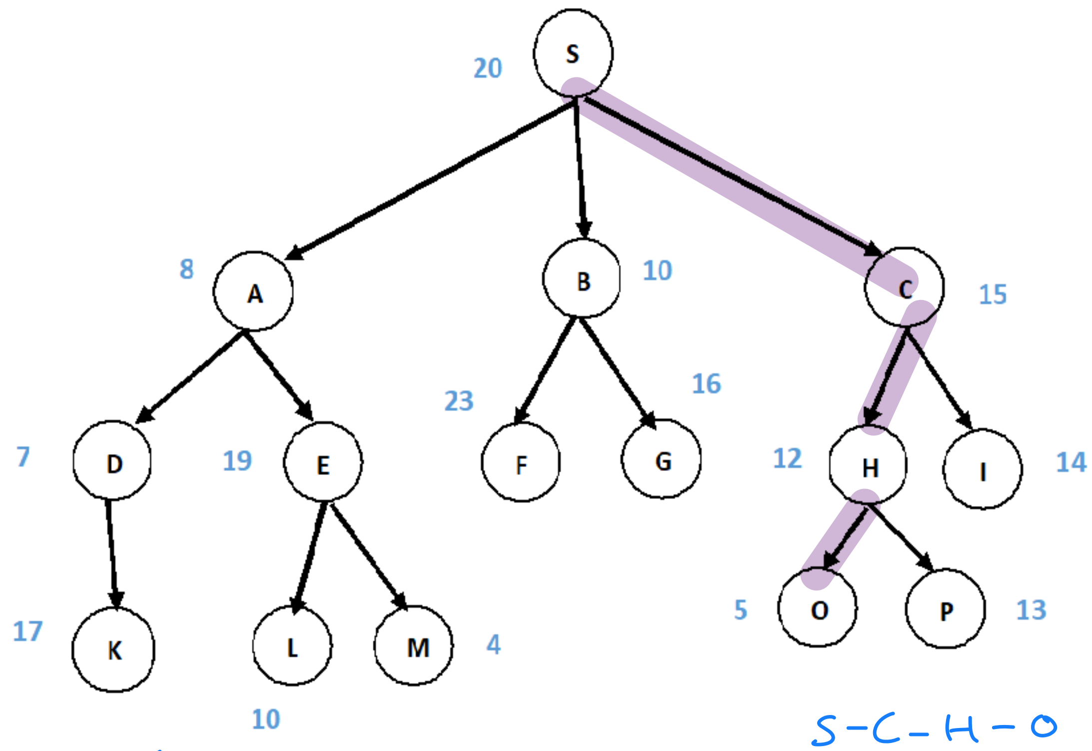
a. Road or Highway
b. Person who are in charge to monitor AV car
c. Weather condition

4. Referring to Table 1, what is the Sensor (C) for the case study? (2 marks)
   a. GPS
   b. Breaking system
   c. Traffic light

5. Referring to Table 1, what are the Actuator (D) for the case study? (2 marks)
   a. Making decision based on the environment to manoeuvre through the traffic
   b. Update information and find the shortest path
   c. Able to track the driver's location

QUESTION 4 -CH 4A Search Algorithms

a.Consider the following graph. Find the most cost-effective path from initial state S to reach the goal state O using Best First Search Algorithm.



S-C-H-O

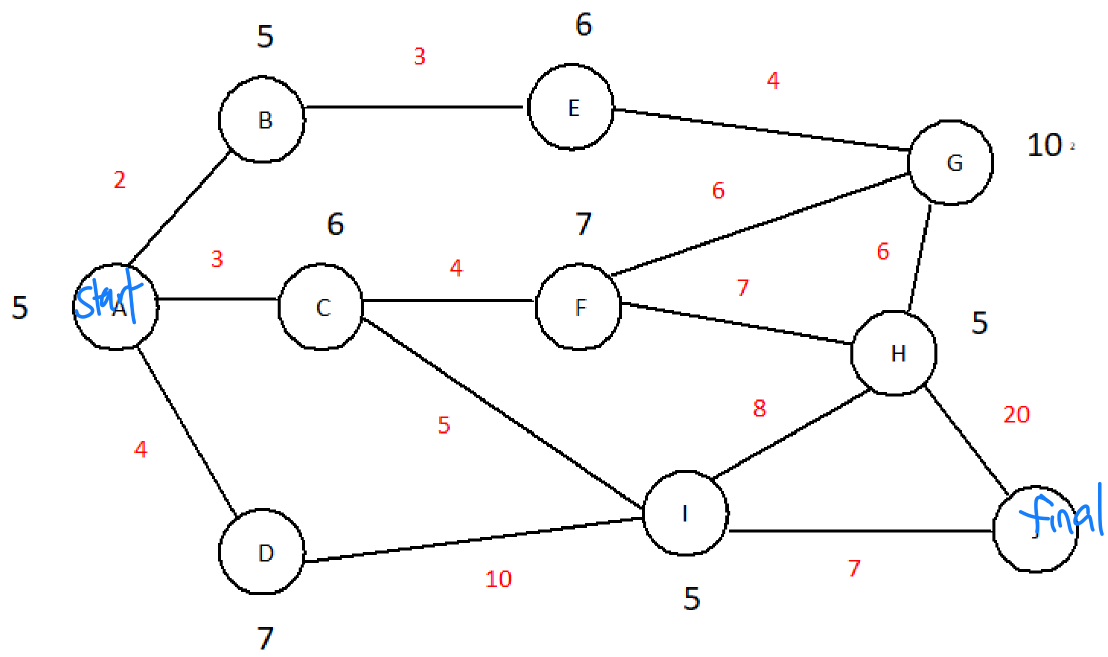
open : [S20] , closed = []
open = [A8, B10, C15] , closed = [S20]
open = [D7, B10, C15, E19] , closed = [S20, A8]
open = [B10, C15, K17, E19] , closed = [S20, A8, D7]
open = [C15, G16, K17, E19, F23] , closed = [S20, A8, D7, B10]

open = [H12, I14, G16, K17, E19, F23], closed = [S20, A8, D7, B10, C15]

open = [O5, P13, I14, G16, K17, E19, F23], closed = [S20, A8, D7, B10, C15, H12]

→ goal is found.

b. Consider the following directed graph.

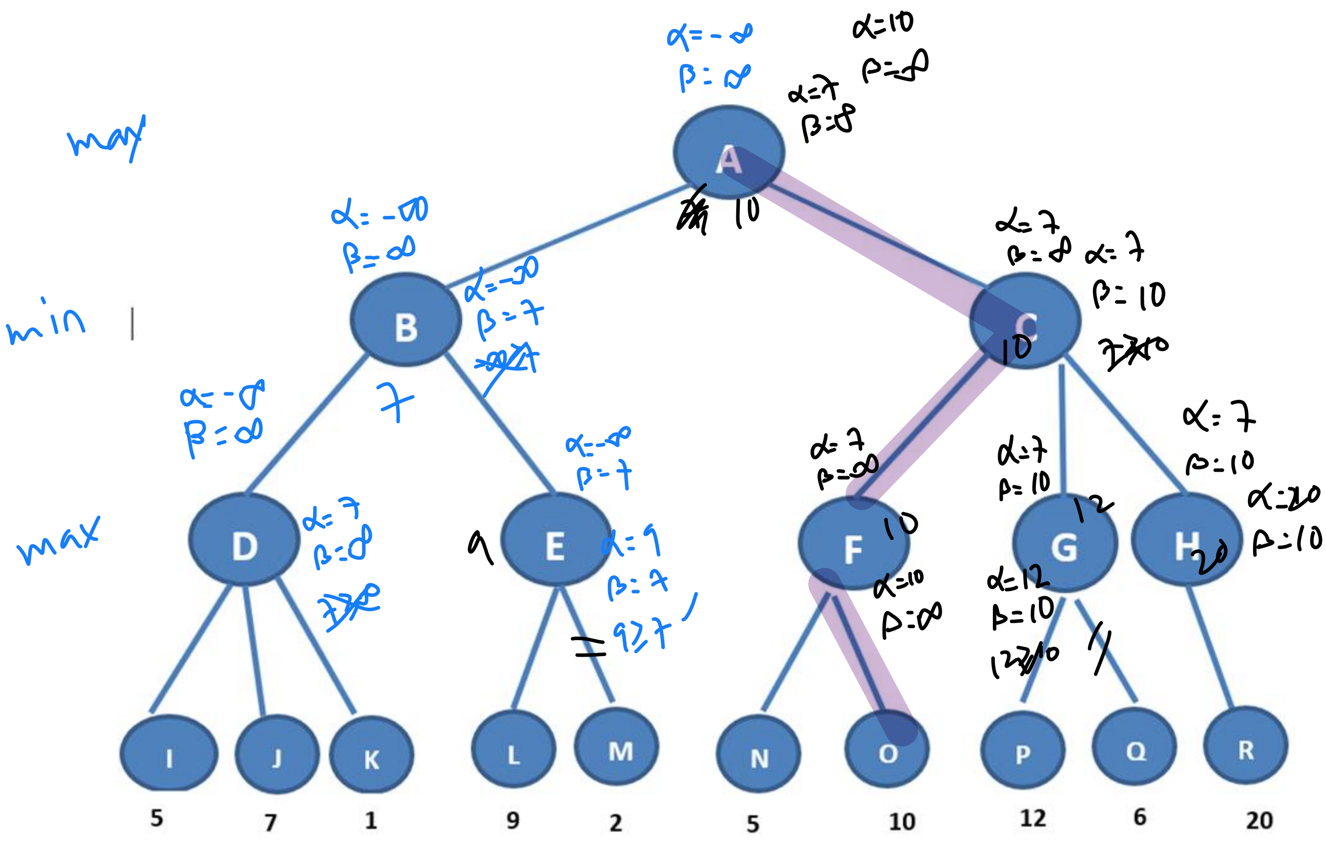


The numbers written on edges represent the distance between the nodes.
The numbers written on nodes represent the heuristic value.
Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.

Node	g(n)	h(n)	f(n) = g(n) + h(n)	prev node
A	0	5	5	
B	2	5	7	A
C	3	6	9	A
D	4	7	11	A
E	5	6	11	B
F	7	7	14	C
G	9	10	19	E
H	14	5	19	F
I	8	5	13 13	C
J	15	0	15	I

A-C-I-J = 3+5+7 = 15

QUESTION 5 CH 4C HEURISTIC IN GAMES



You are required to apply Minimax with Alpha Beta pruning procedure to the search tree in the figure above. Analyze and calculate the values for the following questions. [10m]

a. Calculate the values for nodes A , B, C, D, E, F, G and H?

b. Examine and select the nodes that will not be evaluated. m, Q

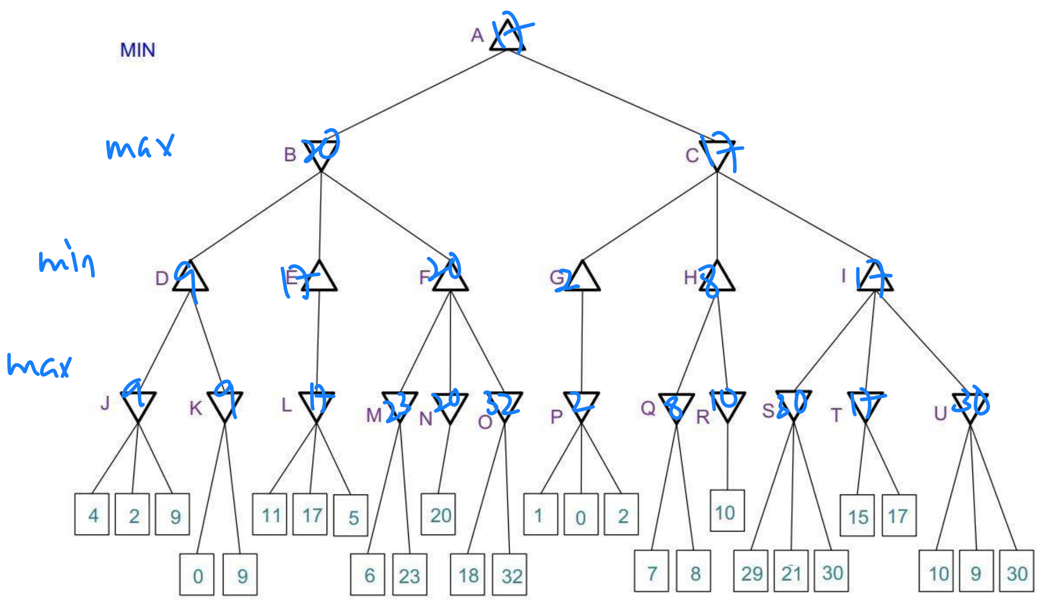
c. Calculate the optimal value for MAX to win the game? 10

d. Examine and select all the nodes in the optimal path for MAX to win the game.

A, C, F, O

QUESTION 6 -MINIMAX

Complete all states (A – U) in the following tree using MINIMAX without Alpha Beta pruning. (10 m)



QUESTION 7

CH7 Machine Learning

Figure A shows COVID-19 attributes and their classification results (assume COVID-19 positive case be 1 and the negative case is 0). Below are the detailed attributes.

Attributes:

1. LC - Location

2. CO- Country

3. GE – Gender

4- AG- Age

5- CH-Cough

6- CP- Chest Pain

7- FE- Fever

LC	CO	GE	AG	CH	CP	FE	TESTED_POSITIVE	PREDICTED_POSITIVE
104	8	1	66	1	0	1	1	0
101	7	0	56	0	1	0	1	0
137	6	1	46	0	1	0	1	1
116	9	0	60	1	0	0	0	0
116	6	1	58	0	0	0	0	0
23	6	0	44	0	1	0	1	1
105	7	1	34	0	1	0	1	1
13	7	1	37	1	0	0	0	1
13	8	1	39	1	0	0	1	0
13	8	1	56	1	0	0	1	0
13	9	0	18	1	0	0	1	1
13	9	0	32	1	0	0	0	1
100	7	1	37	1	0	0	0	1
135	7	1	51	0	1	0	0	0
105	7	1	57	0	1	1	1	1
53	6	1	56	1	0	0	0	0
53	6	1	50	1	0	0	0	0
71	8	0	52	0	1	0	0	0
67	9	1	33	1	0	0	0	1
67	8	1	40	1	0	0	1	1

Predict

Actual	1	0
1	6 TP	4 FN
0	4 FP	6 TN

Based on the dataset given in Figure A, answer the following questions:

a) Quantify the amount of True Positive (TP) 6, True Negative (TN) 6, False Positive (FP) 4 and False Negative (FN) 4 for the classification result **[4 m]**

b) Using your answers in a), calculate the accuracy, precision, and recall for this classification **[6 m]**

accuracy = $TP + TN / (TP + TN + FP + FN)$

$= 12 / 20$

$= 0.6$

precision $= TP / (TP + FP)$

$= 6/10$

$= 0.6$

recall $= TP / (TP + FN)$

$= 6/10$

$= 0.6$